Diffusion Quick Assessment

1. Define diffusion:

2. Diffusion can happen in: (circle the correct answers)

   solids **liquids** **gases**

3. Explain your answer to question 2 using the particle model.

4. A teacher set up the following apparatus:



   a) Describe what would happen to the bromine gas particles as the card in between the two glass jars was removed:

   b) Why might the diffusion of this gas actually be slower than expected?

   c) How could the teacher speed up the rate of diffusion in this experiment?

- ☐ I can define diffusion.
- ☐ I can explain how diffusion works in terms of particles.
- ☐ I can suggest factors that could affect the rate of diffusion.
- ☐ I can predict the movement of particles.

Diffusion Quick Assessment **Answers**

1. Diffusion is:

   - **the movement of liquid or gas particles;**
   - **from an area of high concentration;**
   - **to an area of low concentration;**
   - **down a concentration gradient;**
   - **requiring no energy/it is a passive process.**

2. Diffusion can happen in **liquids and gases.**

3. **Diffusion can happen in liquids and gases because the particles are free to move, whereas in a solid the particles are fixed and can only vibrate on the spot.**

4. A teacher set up the following apparatus:

   a) **The bromine gas particles would diffuse from a high to low concentration and so move upwards.**

   b) **The diffusion of this gas could be slower than expected due to any air particles colliding with the bromine gas particles and sending them off in another direction. This random movement of gases is known as Brownian motion.**

   c) **The teacher could speed up the rate of diffusion in this experiment by increasing the amount of bromine gas used to create a steeper concentration gradient or increasing the temperature. (Size of molecule and state of matter are not applicable in this experiment as bromine gas is being used.)**

Diffusion Quick Assessment **Teacher Feedback**

Effort: 1 2 3 4 5

You can define diffusion using **some** keywords.	You can define diffusion using **most** keywords.	You can define diffusion using **all** keywords.
You can **identify** that diffusion can only happen in liquids and gases.	You can **explain** why diffusion can only happen in liquids and gases using the particle model.	You can explain why diffusion can only happen in liquids and gases and **compare** this to solids using the particle model.
You can **state** that gas particles move quickly.	You can **describe** the random movement of gases due to collisions.	You can **apply** Brownian motion to the diffusion of gas particles in a particular experiment.
You can list **one** factor that would increase the rate of diffusion.	You can list **all** factors that would increase the rate of diffusion.	You can **select** which factors would increase the rate of diffusion in a particular experiment.

Next Steps:

